

GEOLOGY FOR ENGINEERS AND SCIENCES LAB (GLE 203)

List of Experiments

1. Study of Physical properties of minerals
2. Identification of rocks forming silicate minerals
3. Identification of ore minerals
4. Recognition of Igneous rocks
5. Recognition of Sedimentary rocks
6. Recognition of Metamorphic rocks
7. Triangular method for construction of contours
8. Construction of contours and identification of structures
9. Geological cross sections and study of geological maps.
10. Study of geological structures and out crops patterns and land forms
11. Outcrop Patterns of Folded Rocks

Objective of Geology Lab

- ❖ Identification of Minerals and rocks.
- ❖ To understand the use of topographic map and contour maps.
- ❖ To apply geologic concepts and approaches Petroleum geology projects.
- ❖ To identify and characterize intact rock/rock mass properties.

Things to Do

1. Be on time. At the start of the lab period, there will be a short introduction to the experiment you will perform that day. It is unfair to your partner and to others in the lab if you are not up to speed when the work begins.
2. Inform the instructor and/or TA if there is a problem. You will have their immediate attention if you have cut yourself (even if you consider it minor), if something broke and needs cleaning up, or if you are on fire.
3. Be aware of all the safety devices
4. Keep clutter to a minimum.
5. Wash your hands before you leave the lab for the day.
6. Be aware of others in the lab.
7. Bring your lab notebook and an open mind to every lab meeting.

Things Not to Do

1. Do not eat, drink, chew gum, smoke or apply cosmetics in the lab. Just being in lab makes your hands dirtier than you can imagine.
2. Do not put pieces of lab equipment in your mouth.
Do not break Samples in pieces.

Experiment :- 1

Objective:- Study of Physical properties of minerals

Theory:- Earth is made up of minerals that are the constituents of rocks. Mineral specimens are usually identified by determining their physical properties.

Colour: Although the colour of some minerals, such as azurite, is quite distinctive, other minerals, such as quartz, occur in a variety of colours. Also there are many white minerals. Hence colour is frequently NOT a useful diagnostic property.

Streak: Streak is the colour of the powdered mineral. It is a useful diagnostic property for many coloured minerals — especially those with a metallic lustre. It is found by rubbing the specimen on a piece of unglazed tile, or streak plate.

Lustre: The lustre of a mineral is the way its surface shines when held up to the light. Lustre is a property distinct from colour. There are many ways of classifying and describing lustre, but the following system is adequate:

Vitreous — the mineral shines like glass — e.g. quartz, diamond

Metallic — the mineral shines like the surface of a metal — e.g. pyrite, galena Earthy (dull) – the mineral does not shine at all — e.g. kaolinite

Hardness: The hardness of any mineral can be assigned a number between 1 and 10, on Moh's Scale of Hardness. The instruments used to determine the hardness of a mineral specimen are (in order of increasing hardness) a finger-nail, copper coin, knife blade and a quartz crystal.

The table below lists the minerals that define **Moh's Scale of Hardness**, and gives the relative harnesses of the test items named above.

	Defining Mineral	Test Item
1	Talc	
2	Gypsum	
3	Calcite	Finger nail ~2.5
4	Fluorite	Copper coin ~3.5
5	Apatite	Knife blade ~5.5
6	Orthoclase	
7	Quartz	Quartz crystal 7
8	Topaz	
9	Corundum	
10	Diamond	

Density: It is not usual to measure the actual densities (relative to water = 1) of specimens; however, minerals should be classified according to whether they are light, medium or heavy. This can be done by holding similar-sized specimens of two different minerals in your hands, and comparing their weights.

Cleavage: When a piece of a mineral is dropped or struck, it may tend to break so that flat, shiny surfaces are formed.

Minerals, or individual surfaces, that do not cleave to form flat faces are said to show fracture. Cleavage is a diagnostic property for identification of minerals, but the cleavage of an

actual specimen is not always easy to determine. Many specimens do not show the expected cleavage characteristics.

Magnetism: Some minerals that contain iron are magnetic. Magnetite is strongly magnetic, and will be attracted by a magnet. Other iron-bearing minerals such as ilmenite sand size particles.

Reaction to dilute Hydrochloric Acid: Some minerals especially carbonates, effervesce when a drop of dilute hydrochloric acid is placed on them. This is useful diagnostic test for calcite and a white mineral which is not easily distinguished.

Apparatus and components required: Streak plate, minerals, hand lances

EXPERIMENT :-2

Objective:- Identification of rocks forming silicate minerals

Theory:- Although about 4000 minerals are known to exist, only about 8 of them are common. These common rock forming minerals are the major constituents of igneous, sedimentary and metamorphic rocks. They constitute more than 99% of Earth's crust. The following table lists the common rock-forming minerals.

Mineral	Group	SIGNIFICANT DIAGNOSTIC PROPERTIES				
		H	Colour	Lustre	Cleavage	Other properties
Quartz	Silicate	7	Clear when pure. Impurities cause many colour variations.	Vitreous	None	Hexagonal crystals
Feldspar group: Orthoclase Plagioclase	Silicate	6	Orthoclase: pink, cream Plagioclase: white, grey	Vitreous	2 at ~90°	
Biotite	Silicate	2.5	Black	Vitreous: sometimes appears metallic	1	Thin sheets are flexible and elastic
Muscovite	Silicate	2.5	White or clear		1	
Amphibole (e.g. Hornblende)	Silicate	5.5	Black	Vitreous	2 at 120°	Often confused with Pyroxene
Olivine	Silicate	6.5	Green	Vitreous	none	Small green crystals. Often enclosed in a basalt volcanic 'bomb'.
Pyroxene (e.g. Augite)	Silicate	5.5	Black	Vitreous	2 at 87° & 93°	Often confused with Amphibole
Calcite	Carbonate	3	White or clear	Vitreous	3 not at 90°	Effervesces with acid
Clays (e.g. Kaolinite)	Silicate	2.5	White	Dull	None	Very powdery

Properties:- Properties Such as Hardness, Density and cleavage are often impossible to determine in these specimens.

Apparatus and components required: Streak plate, silicate minerals, hand lances

EXPERIMENT :-3

Objective:- Identification of ore forming minerals

Theory:- Apart from rock forming mineral ore forming minerals are also present which are of interest of industry. The common ore forming minerals can be easily identified by their colour, lustre and streak

Mineral	Composition	SIGNIFICANT DIAGNOSTIC PROPERTIES				
		H	Colour	Lustre	Streak	Other properties
Galena	Lead sulphide (PbS)	2.5	Grey	Metallic	Lead-grey	Very high density, 3 cleavage planes at 90°
Chalcopyrite	Copper iron sulphide (CuFeS ₂)	3.5	Greenish-gold or many colours (iridescent)	Metallic	Greenish black	Iridescent specimens are known as 'peacock ore'.
Malachite	Copper carbonate [Cu ₂ CO ₃ (OH) ₂]	3.5	Green	Vitreous or dull	Green	Green colour is diagnostic.
Sphalerite	Zinc sulphide (ZnS)	3.5	Brown/black	Metallic	Brown	Dodecahedral cleavage (6 planes of cleavage)
Bauxite	Mixture of aluminium hydroxides	2	Brown	Dull	Brown	Consists of round nodules (i.e. pisolitic). Easily recognised.
Haematite	Iron oxide (Fe ₂ O ₃)	6	Reddish brown to black	Usually dull	Reddish brown	Appearance of mineral varies. Streak is diagnostic.
Magnetite	Iron oxide (Fe ₃ O ₄)	6	Black	Metallic	Black	Strongly magnetic.

Apparatus and components required: Streak plate, ore minerals, hand lances

EXPERIMENT No.:- 4

OBJECTIVES:-

- Recognizing igneous rock characteristics.
- Classifying different igneous rocks.

Igneous Rocks:

- Granite
- Basalt
- Gabbro
- Dolerite
- Rhyolite

MATERIALS:

- Identification table
- Rocks

Theory:-

Rocks hold records of the earth's history. When we get a piece of rock in lab it is important to identify type whether they belongs to igneous, sedimentary or metamorphic. An igneous rock provides information about magma type, depth of formation and crystallization of minerals.

PROCEDURE:-

Identification of different mineral phases their shape, size and physical properties of individual mineral. Colour of rock will help to classify and grain size will provide information about their depth of origin.

Apparatus and components required: Streak plate, igneous rocks, hand lances

EXPERIMENT No.:- 5

OBJECTIVES:-

- Recognizing sedimentary rock characteristics.
- Classifying different sedimentary rocks.

Sedimentary Rock:

- i. Sandstone
- ii. Conglomerate
- iii. Blake shale
- iv. Grey Shale
- v. Limestone

MATERIAS:

- iii. Identification table
- iv. Rocks

Theory:-

A sedimentary rock display the surficial process and geological. A piece of sedimentary rock can provide us information about surficial processes, weathering condition, depositional history and fossil records.

PROCEDURE:-

Identification of different mineral phases their shape, size and physical properties of individual mineral. Colour of rock will help to classify and grain size and structures will provide information about their transportation history and depositional environment.

Apparatus and components required: Streak plate, sedimentary rocks, hand lances

EXPERIMENT No.:- 6

OBJECTIVES:-

- Recognizing Metamorphic rock characteristics.
- Classifying different Metamorphic rocks.

Sedimentary Rock:

- i. Slate
- ii. Marble
- iii. Schist
- iv. Gneiss
- v. Amphibolite

MATERIAS:

- i. Identification table
- ii. Rocks

Theory:-

A Metamorphic rock display the transformation from its protoliths and also provide information

about geological condition. A piece of metamorphic rock can provide us information about pressure and temperature condition of their formation and also provide information about tectonic condition from set of mineral assemblages.

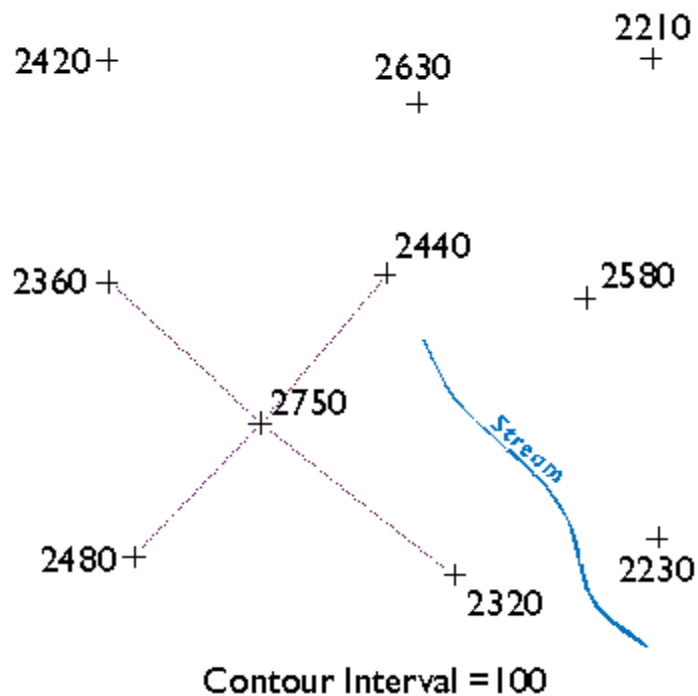
PROCEDURE:-

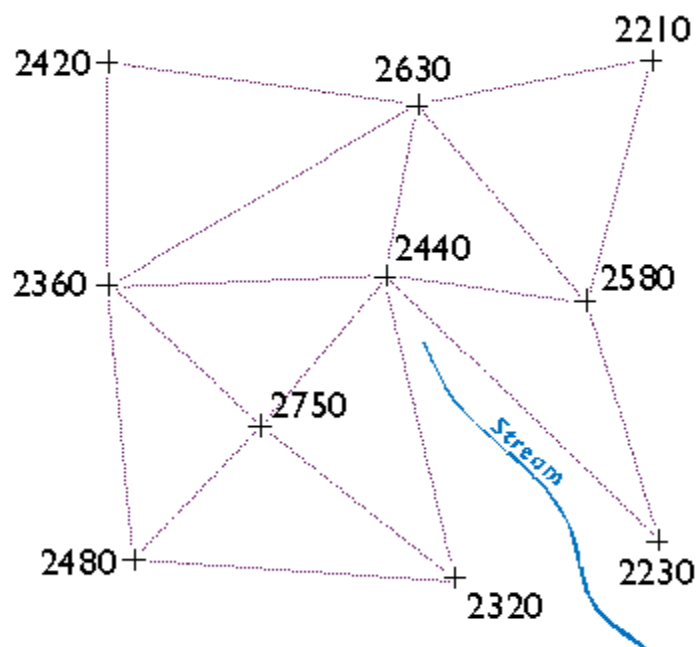
Identification of different mineral phases their shape, size and physical properties of individual mineral. Structures mineral assemblages help to classify and infer PT condition of their formation.

Apparatus and components required: Streak plate, metamorphic rocks, hand lances

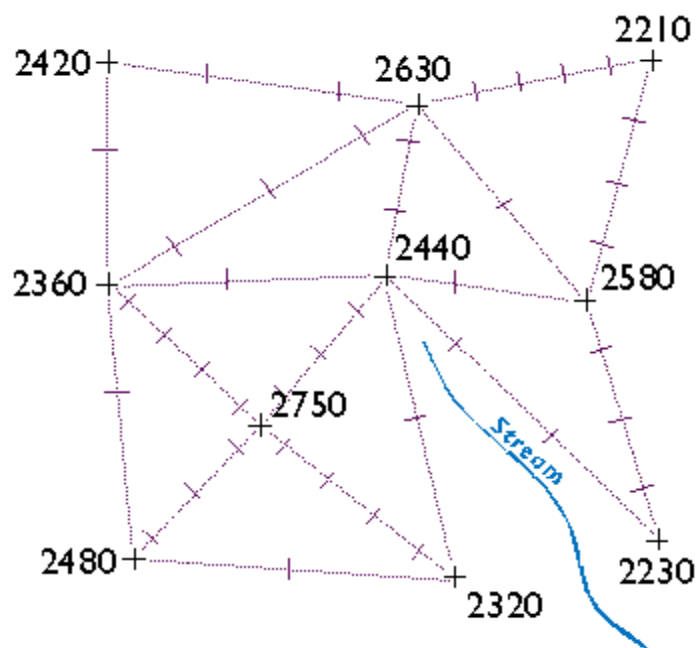
EXPERIMENT No.:-7

Objective: Triangulation method for construction of contour

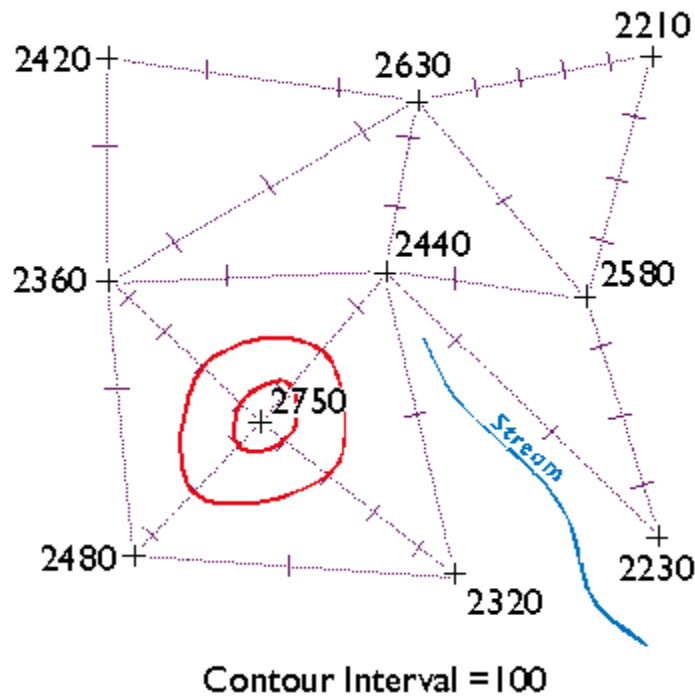




Contour Interval = 100



Contour Interval = 100



Procedure:

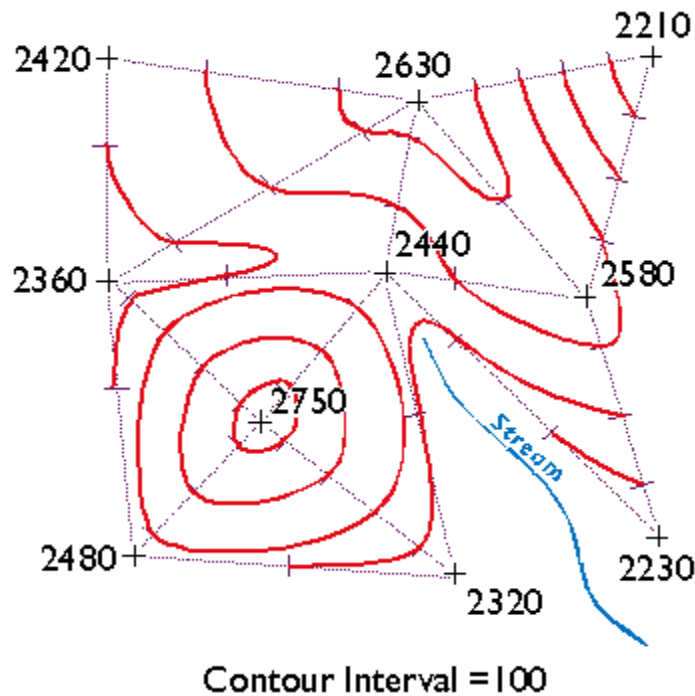
Starting at the highest elevation, draw straight lines to the nearest neighbouring spot elevations. Once all the neighbouring points have been connected with the highest point process has to repeat for the second highest elevation. Taking care not to draw triangles across the stream, continue until the surface is completely triangulated.

The result is a **triangulated irregular network (TIN)**. A TIN is a vector representation of a continuous surface that consists entirely of triangular facets. A useful characteristic of TINs is that each triangular facet has a single slope degree and direction. With a little imagination and practice, you can visualize the underlying surface from the TIN even without drawing contours.

Now draw ticks to mark the points at which elevation contours intersect each triangle side. For instance, see the triangle side that connects the spot elevations 2360 and 2480 in the lower left corner of Figure. One tick mark is drawn on the triangle where a contour representing elevation 2400 intersects. Now find the two spot elevations, 2480 and 2750, in the same lower left corner. Note that three tick marks are placed where contours representing elevations 2500, 2600, and 2700 intersect.

Finally, draw contour lines. Working downslope from the highest elevation, thread contours through ticks of equal value. Move to the next highest elevation when the surface seems ambiguous. Keep in mind the following characteristics of contour lines (Rabenhorst and McDermott, 1989):

- Contours should **always point upstream** in valleys
- Contours should **always point downridge** along ridges
- Adjacent contours should **always be sequential or equivalent**
- Contours should **never split into two**
- Contours should **never cross or loop**
- Contours should **never spiral**
- Contours should **never stop in the middle of a map**



Apparatus and components required: pencil, elevation data points, diagonal scale, calculator

EXPERIMENT No- 8

Objective: Construction of contours and identification of structures

- Know the definition of a structure contour.
- Be able to hand contour data.
- Be able to calculate surface traits (e.g. local strike and dip) from structure contours.
- Construct map pattern of a contact from structure and topographic contours.
- Be able to use Surfer (or equivalent software) to contour data.
- Be able to interpret the structure contours geologically.

Structure contour maps are basically the same as topographic maps, where the surface being mapped is some geologic surface, often in the subsurface. Geologic surfaces commonly mapped in this way are:

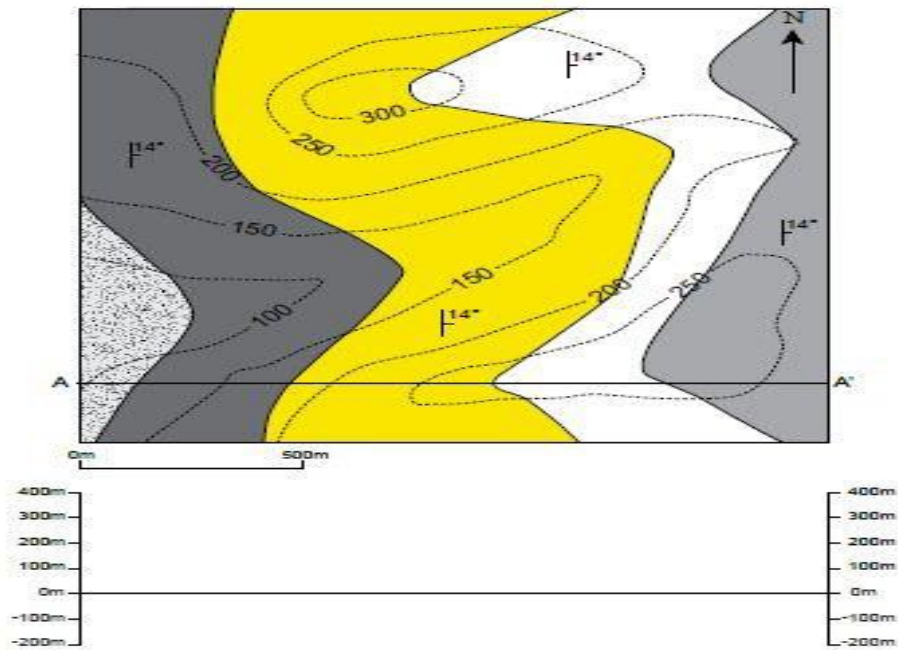
- a stratigraphic horizon.
- a fault surface.
- top of the groundwater table.
- a seismic reflection surface.

However, one can get much more sophisticated and contour for example the amount of offset on a fault surface (which shows a very interesting pattern). As we will see in later labs one can also map the density of point data.

There are some differences between standard topographic maps and structural contour maps. Your surface does not have to be continuous everywhere, but can be truncated. For example the surface created by the top of a sill can be truncated by a fault.

You can also contour and 'map' other features than a physical surface. In this case the z axis instead of being elevation can be any physical/chemical parameter. For example, you can contour geophysical parameters such as the acceleration of gravity, or the strength of a magnetic field.

When working with contaminant plumes the z parameter is the concentration of some contaminant.

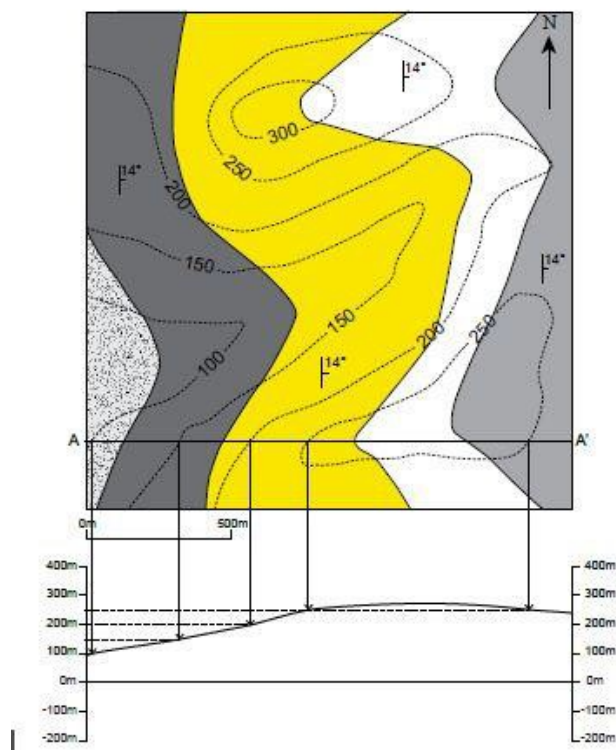


Step 2:

Draw axes of an appropriate scale with the topographic values. Unless there is a reason to do otherwise, draw a true-scale section.

Step 3:

Transfer the topographic information from the map to the section. Project the height of each topographic contour, where it crosses the line of section, on to the section and draw in the topography.

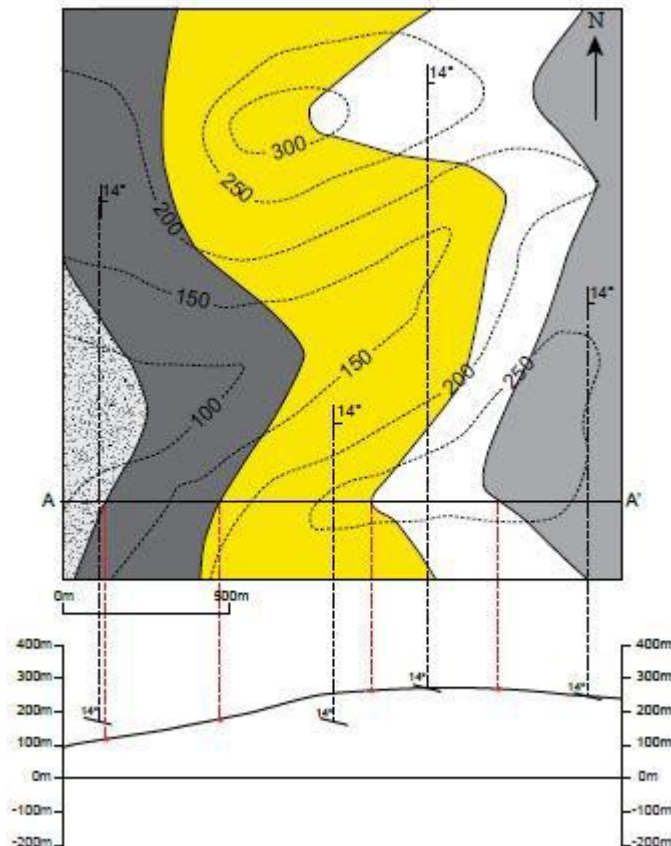


Step 4:

Transfer the lithological boundaries, faults etc onto the cross section in the same way.

Step 5:

Transfer bedding readings on to the section, correcting for apparent dip if necessary (see figure). Plot the readings at the height at which they occur, so where a reading is extrapolated from a greater or lesser height than the topography of the cross section plot it above or below the topography as appropriate.

**Step 6:**

Using the bedding readings as a guide, drawing the lithological boundaries both above and below the surface. Geology extended above the topography is shown by dashed lines. When drawing the section always consider what is geologically reasonable behavior for the layers e.g. sudden changes in a unit's thickness or dip should be justifiable.

Apparatus and components required: geological map with topography, pencil, diagonal scale

EXPERIMENT No.:- 10

Objective:- Study of geological structures and out crops patterns and land forms

Theory:- If the Earth's surface was flat, if there was no topography, then geological maps would be simple. They would be a direct reflection of the underlying geology. However, topography interacts with the geology to produce more complex but predictable patterns.

Horizontal and vertical strata:

Horizontal and vertical strata are the most straightforward to interpret on a map. Horizontal outcrops will always follow the topographic contours (figure 2a), whilst vertical layers will always form straight lines (figure 2b).

Dipping strata

Dipping layers interact with the topography in predictable ways. In valleys, beds will appear to ‘vee’ either up or down the valley in the direction of dip (figure 2c and 2d). This is because the valley side acts as an approximate cross section—not always helpful on small scale maps but on large scale maps and in the field this is a useful aid in interpretation.

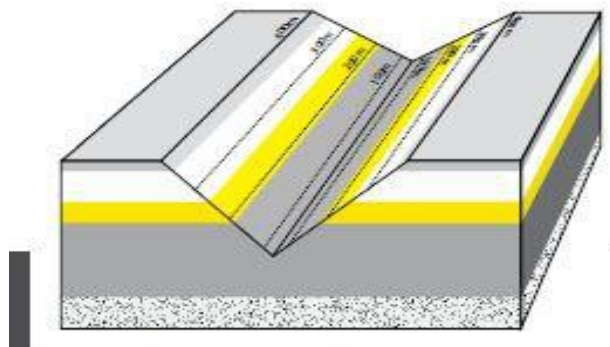


Figure 2a: Horizontal beds

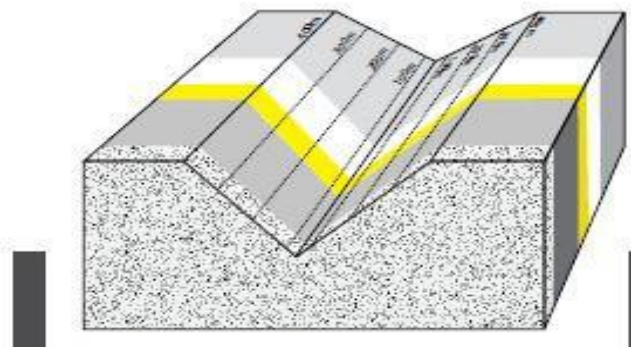


Figure 2b: Vertical beds

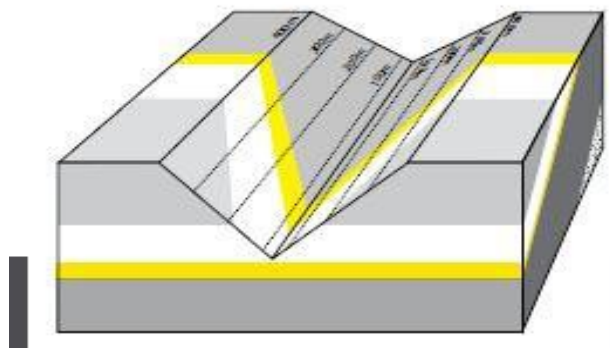


Figure 2c: Beds dipping towards viewer

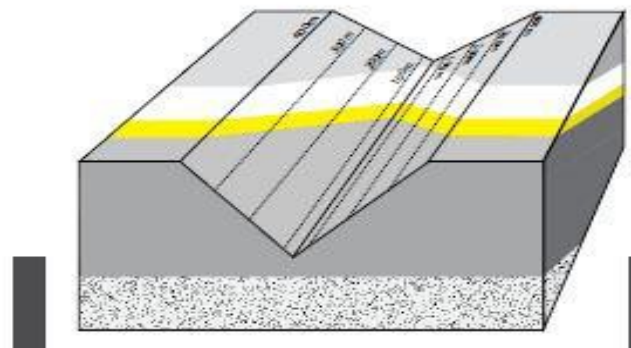


Figure 2d: Beds dipping away from viewer

➤ Outcrop, structure and age relationships

A. Folding of rocks

a. Layer cake relations

- (1) oldest on bottom, youngest on top

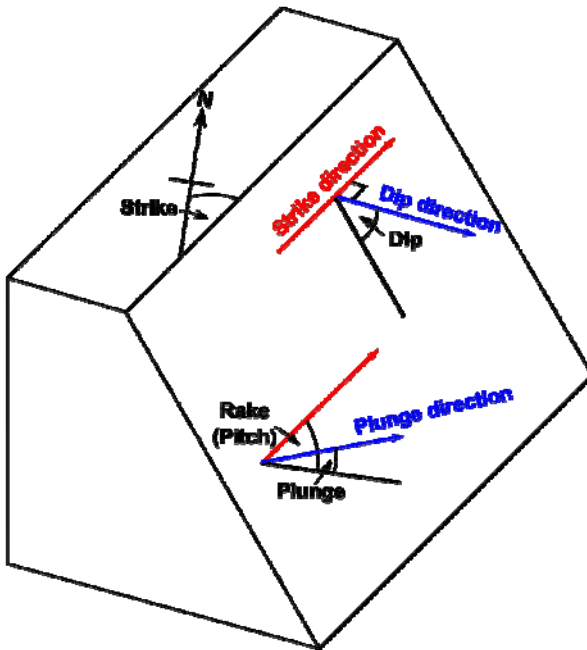
b. Fold Types

- (1) Anticlines-upfolded forms, results in older rocks becoming enclosed within younger strata
- (2) synclines-downfolded forms, results in younger rocks becoming enclosed within older strata.
- (3) symmetrical folds- both limbs of the fold dipping at same angle away from fold axis
- (4) asymmetrical folds- both limbs of the fold not dipping at same angle away from fold axis
- (5) overturned folds- condition in which one limb of fold has been tilted beyond vertical
- (6) plunging folds- axis of fold is tilted

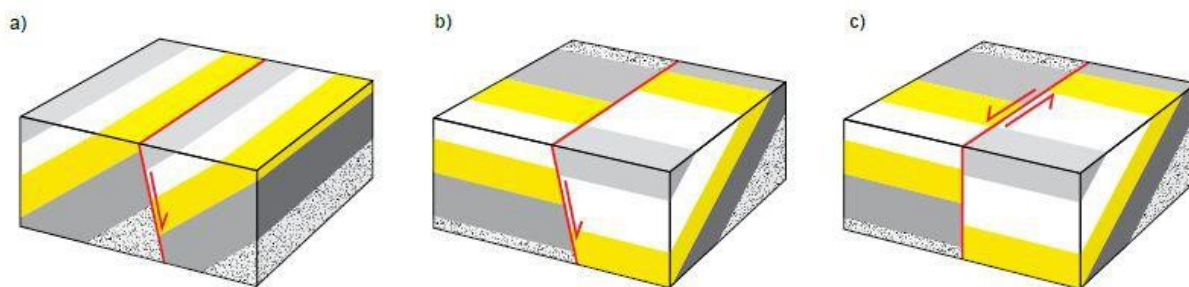
- (7) Domes- more or less circular equivalent of anticline, oldest rocks exposed in center of dome
- (8) Structural Basin- more or less circular equivalent of syncline, youngest rocks exposed in center of dome (not to be confused with depositional basin)

c. Outcrops Patterns Associated with Folded Rocks

- (1) As rocks are folded, and subsequently subjected to erosion, regular patterns become evident in relation to type of rock that outcrops and age of the rock that outcrops in an area of folded strata. In essence, erosion exposes the interiors of the folds
- (2) Non-plunging Folds- axis of fold is horizontal, results in parallel bands of dipping strata about the fold axis
 - (a) anticlines- oldest strata exposed along fold axis
 - (b) synclines- youngest strata exposed along fold axis
- (3) plunging Folds- axis of fold is tilted, results in alternating V-shaped bands of dipping strata oriented about the fold axis.
 - (a) anticlines- oldest strata exposed in the center of the V, V points in direction of plunge of fold axis
 - (b) syncline- youngest strata exposed in the center of the V, V points in opposite direction of plunge of fold axis.
- (4) Doubly Plunging Folds- fold axis is plunging in two opposite directions, results in a flattened oval pattern, or a double V-shaped pattern <<<◇>>>.
 - (a) anticlines- oldest strata exposed in center of flattened oval
 - (b) synclines- youngest strata exposed in center of flattened oval.



Faults on maps:- Faults can occur at any angle with respect to bedding and so the outcrop patterns produced are not unique to any one fault type. Figure a and b show two potential outcrop patterns for anormal fault.



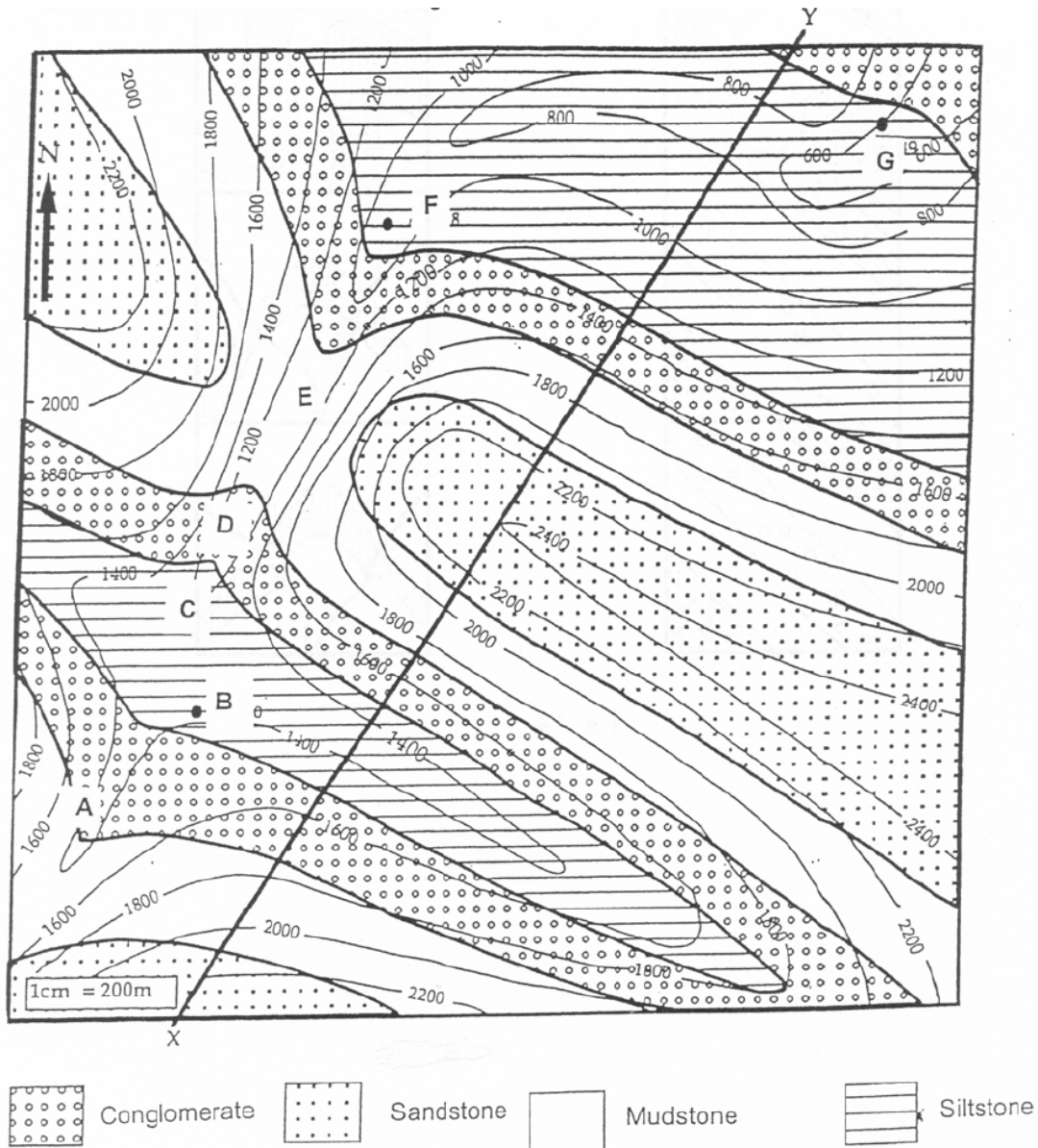
Fold on maps:- Figure shows the different outcrop patterns for folds. The limbs of the folds form a repeating pattern on either side of the axial plane. For anticlines, the oldest rocks are in core of the fold and the rocks get younger away from the axial trace (figure a). For synclines, the youngest rocks are in the core and the rocks get older away from the axial trace (figure b).

Apparatus and components required: lithological map with contour, pencil, diagonal scale

EXPERIMENT No.:- 11

Objective: Outcrop Patterns of Folded Rocks

Theory: A cylindrical fold is outlined by the movement of a straight line, parallel to its original orientation, through space. The folded surface may be regarded as a set of many parallel straight lines. The orientation of any of these lines is the fold axis. The axis of a cylindrical fold is defined as the line which may be moved parallel to itself to form the fold surface. Where a single fold affects a series of beds, the surface containing the hinge lines of all the folded surfaces is called the axial surface. If the axial surface is a plane, it is called the axial plane. The line along which the axial surface intersects the land surface is the axial trace. The plunge of a fold is the orientation of the fold axis. The hinge line of a fold is parallel to the axis and is situated where the curvature of the fold is greatest. The hinge line usually provides the easiest place in which to measure the fold axis and is often wrongly referred to as the fold axis.



Apparatus and components required: Worksheet, Pen or pencil